Giannina Guzmán Caloca

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Education

University of Maryland
Degree: M.S., Astronomy — Graduation Date: January 2023
Degree: Ph.D., Astronomy — Expected Defense: Spring 2027

Dissertation: "Of sub-Neptunes and GEMS: a broad look into the atmospheres of planets with no Solar System analogues" First, I explore the limitations of our atmospheric grid retrieval tools when taking into account computational performance. Then, I use retrievals in the extensive analysis of several targets observed with JWST and HST transmission spectroscopy that have no Solar System analogues, such as Giant exoplanets around M-dwarf Stars (GEMS) and sub-Neptunes.

Villanova University
Degree: Bachelor of Science — Graduation Date: May 2019
Major: Astrophysics and Planetary Science
Minor(s): Communication, Physics

Professional Work Experience

NASA Post-baccalaureate — **September 2019-August 2020**
Developed software for the Planetary Spectrum Generator — Goddard Space Flight Center
Software Engineer — **June 2019-August 2019**
Developed software and helped design telescope automation framework — Lowell Observatory
Software Developer — **August 2018-June 2019**
Developed software for NASA PDART 'sbpy' project — Villanova University
NASA GSFC URAA Intern — **June 2018-August 2018**
Goddard Space Flight Center
Public Observatory Supervisor — **August 2015-May 2017**
Villanova University

Teaching and Mentoring

Mentor for a High School Student — **October 2026-Present**
Mentoring a high school student in Puerto Rico on exoplanet research for her senior capstone project.
NASA Funded Taller Futur@ Astronom@ — **Summer 2024-Present**
Teaches a 2.5 hour intro to Astrobiology lecture in Spanish + designs homework
Teaching Assistant — **August 2020-2023**
ASTR 101 - "Introduction to Astronomy" — University of Maryland
ASTR 220 - "Collisions in Space – The Threat of Asteroid Impacts"
Teaching Assistant — **August 2016-May 2018**
Villanova University

Awards and Honors

International Conference Student Support Award (ICSSA) — **Summer 2026**
ExoSLAM 2023 Summer School Attendee — **Summer 2023**
Jason A. Cardelli Memorial Award for Undergraduate Research — **Spring 2019**
"The Jason A. Cardelli Memorial Award for Undergraduate Research is presented to a graduating Astronomy & Astrophysics major whose body of undergraduate research work exhibits particularly high standards of independence, originality, and quality." (Villanova University Website)

First place in Villanova's 2018 Sigma Xi CRF Poster Symposium **Spring 2018**

Publications

First author

[1] **Guzman Caloca, G**., et al. (2025). GEMS JWST: Hold on to your HATS(-6 b), a sub-solar metallicity planet with water, methane and ammonia in its atmosphere. AJ, 172, 4. DOI: 10.3847/1538-3881/ae9858

[2] **Guzman, G.**, Sion, E., & Godon, P. (2019). FUSE and IUE Spectroscopy of the Prototype Dwarf Nova ERUrsa Majoris during Quiescence. Astronomical Journal. AJ, 158, 99. DOI:10.3847/1538-3881/ab322f

Significant Contributions

[3] Kanodia, S., **et al**. (2026). GEMS JWST: A sub-Solar metallicity atmosphere for giant planet TOI-5293Ab orbiting a rapidly changing M-dwarf. ApJ, 1006, 119. DOI: 10.3847/1538-4357/ae801f

[4] Ashtari, R., **et al.** (2026). GEMS JWST: HATS-75 b, A giant planet with a sub-solar metallicity atmosphere orbiting an M-dwarf. AJ, 171, 294. DOI: 10.3847/1538-3881/ae552c

[5] Cañas, C. I., **et al.** (2026). GEMS JWST: Transmission spectroscopy of TOI-5205b reveals significant stellar contamination and a metal-poor atmosphere. AJ, 171, 260. DOI: 10.3847/1538-3881/ae4976

[6] Bell, T., **et al.** (2022). Eureka!: An End-to-End Pipeline for JWST Time-Series Observations. JOSS, 7(79), 4503, DOI: 10.21105/joss.04503

[7] van Belle, G.T., Collins, M., **Guzman, G.**, Mommert, M. (2020). Improved ASCOM Dome Following. Research Notes of the AAS, 4, 8. DOI:10.3847/2515-5172/abb29b

[8] Mommert, M., **et al.** (2019). sbpy: A Python module for small-body planetary astronomy. JOSS, 4(38), 1426. DOI:10.21105/joss.01426

Others

[9] Ginsburg, A., **et al.** (2019). astroquery: An Astronomical Web-querying Package in Python. AJ, 157, 3. DOI:10.3847/1538-3881/aafc33

[10] Campbell, H., Sheldon, Z., Gibson, J., **Guzman, G.** (2020). Technological and Mediated Identity in American Multisite Churches. Ecclesial Practices, 7, 1. DOI:10.1163/22144417-bja10002

- This publication with significant contribution was a result of a capstone project in my Communication minor, exemplifying my commitment to excellence in any project I'm involved in.

Presentations

Invited Talks

Penn State Center for Exoplanets and Habitable Worlds Seminar **November 2026**
Invitation to present results of publication [1]

Villanova University Colloquium Speaker **October 2026**
Invited to speak on my research and career track to undergraduates as a Villanova alumnus

Max Planck Institute for Astronomy Exocofee invited speaker **October 2026**
Invitation to present results of publication [1]

JWST Weekly Briefings Invited Speaker **October 2026**
Brief overview of JWST research geared towards the JWST

Flight Operations Team

JWST Science Day @ NASA GSFC **August 2026**
Hidden GEMS: the atmospheres of Giant exoplanets around M-dwarf Stars

ESPF Seminar @ STScI **May 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

Carnegie EPL Astronomy Seminar Series **April 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

Exoplanet Seminar @ Goddard Space Flight Center **March 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

Planetary Lunch @ UC Santa Cruz **March 2026**
Of sub-Neptunes and GEMS: A Broad Look into Atmospheres with No Solar System Analogs

Eureka! An End-to-End Pipeline for JWST Time Series Observations **December 2023**
ExoVAST (Virtual Astronomy Talks) seminar talk

EMAC Workshop on Open-Access Exoplanet Modeling & Analysis Tools **February 2023**
Eureka! An End-to-End Pipeline for JWST Time Series Observations

Contributed Talks

Hidden GEMS: The atmosphere [...] M-dwarf Stars **June 2026**
Cool Stars 2023 Tokyo, Japan

ERES X Presenter **June 2025**
Looking at Giant Exoplanets around M-dwarfs (GEMS) with JWST: NIRSpec Transmission Spectroscopy of the Warm Saturn HATS-6 b

AAS 245th Meeting Talk **January 2025**
Looking at Giant Exoplanets around M-dwarfs (GEMS) with JWST: NIRSpec Transmission Spectroscopy of the Warm Saturn HATS-6 b

AAS 245th Meeting Talk **January 2025**
Being Stardust: A Collaboration between Arts and Sciences

Early Career Science Forum Talk **October 2024**
@ NASA GSFC - Looking at Giant Exoplanets around M-dwarfs (GEMS)

Accurate (Exo)planetary Retrievals via High Dimensional Bayesian Samplers **August 2020**
NASA GSFC Planetary Science Division Symposium

Posters

Hidden GEMS: The atmospheres of [...] **June 2026**
Cool Stars 2023 Poster; Tokyo, Japan

Gridtrievals? A Comparative Study of Retrieval Techniques **June 2023**
ExoClimes 2023 Poster; Exeter, U.K.

The Red Thumbs: Growing Plants on Martian Regolith Simulant **January 2019**
AAS Meeting #233 Poster - Education Category

Designing a Python Module for the Calculation of [..] in Comets **January 2019**
AAS Meeting #233 Poster

Digitizing Villanova University's Eclipsing Binary Card Catalogue **January 2018**
AAS Meeting #231 Poster

Community Service

Professional Service

Class Representative — **January 2021-Present**
Graduate Council, University of Maryland Department of Astronomy

Executive Secretary — **December 2025-May 2026**
3 NASA review panels

Prospective Visit Organizer — **Fall 2022-Spring 2024**
Organized the graduate prospective visit for the Department of Astronomy

UMD Astronomy Community and Belonging Committee Member — **Fall 2021-Fall 2024**

Workshops in Applying to Astronomy Graduate School co-lead — **Fall 2023**
Helped establish and lead a workshop series at UMD for astronomy graduate school applications directed at local undergraduates

UMD Astronomy Undergraduate Mentorship Program Mentor — **Fall 2021-Fall 2023**
Mentor for undergraduate students, helps with professional development

AbGradCon2022 Organizing Committee — **Fall 2021-Fall 2022**
Position of leadership: Public and Media Relations (website designed by myself)

Mental Health Task Force — **Spring 2021**
Co-authored the mental health report for UMD's Astronomy Department; based on a dedicated mental health survey done in 2020

Executive Secretary — **April 2020**
1 NASA Review Panel

Outreach

Space Sciences Outreach Cooperative Co-founder and Co-chair — **May 2023-Present**
Co-founder and co-chair of the University of Maryland Space Sciences Outreach Cooperative (SSOC)

AstroAnimation Consultation Scientist — **September 2024**
Helped students from the Maryland Institute College of Art bring the exoplanet GEMS population to life

NASA JWST x Minecraft Education Collaboration Interviewee — **September 2024**
Presenter for a NASA JWST x Minecraft Education collaboration

Being Stardust Workshop @ Library of Congress — **April 2024**
Participant in an art meets astrobiology collaboration outreach workshop

#MathGals Invited Guest — **Fall 2022**
Invited special guest for Pasadena ISD school district #mathgals club

Skype A Scientist Volunteer — **Spring 2020-Fall 2020**

GSMI Cientifico Latino Volunteer — **August 2020-May 2021**
Student Mentor

The Superlative — **January 2016-May 2019**
Position of leadership (2016-2019): Public Relations and media representative

Villanova Astronomical Society — **August 2015-May 2019**
Position of leadership (2017-2018): Treasurer

All Hands-On Science — **May 2017-May 2018**
Volunteering

Awarded Observation Proposals and Collaborations

Target lead: JWST (Cycle 2 GO #3171; 132.39 hrs)
Led the analysis of one of the survey targets in the program "Red Dwarfs and the Seven Giants: First Insights into the Atmospheres of Giant Exoplanets around M-dwarf Stars"

Target lead: HST (Cycle 30 #17192; 116 orbits)
Led the analysis of one of the survey targets in the program "The SPACE Program: a Sub-neptune Planetary Atmosphere Characterization Experiment"
Co-I: JWST (Cycle 4 GO #7683; 22.6 hrs)
A Parched Giant orbiting a Red Dwarf: Fact or Fiction?
Co-I: JWST (Cycle 3 GO #6284; 13.68 hrs)
Searching for Signatures of Surface-Atmosphere Interaction on a Small Planet in its Magma Era
Co-I: JWST (Cycle 3 GO #5959; 129.96 hrs)
KRONOS: Keys to Revealing the Origin and Nature Of sub-neptune Systems

Press Releases and Media

University of Maryland: ***New Study Detects Ammonia and Unexpected Chill on a Distant Giant Planet***
UMD CMNS Press release for publication [1]: link to press release
Maryland Today Press release for publication [1]: link to press release
Telemundo Mujeres Imparables Interview
Interview with Latin American broadcasting agency Telemundo: link to social media post
Villanova University: ***Mars: fertile ground?***
Villanova University article highlighting my undergraduate research: link to article
WHYY Radio Interview: ***Mars bar won't have beer, but dandelion wine is a possibility***
Radio Interview highlighting my undergraduate research: link to article

Skills

Computational
Python, C, Unix, html, software development (design, version control, testing, documentation)
Professional
Google Suite (Google docs, slides, sheets, classroom, sites, etc.), Microsoft Suite (Word, PowerPoint, Excel), public speaking, web design, organization leadership
Personal
Native Spanish speaker